

PERSON AND ACTIVITY DETECTION WITH GEO-COORDINATE MAPPING USING COMPUTER VISION



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CERTIFICATE OF APPROVAL

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ABSTRACT

This project presents the design and implementation of a real-time **Hand Gesture Recognition System** using classical computer vision techniques to enable touchless human–computer interaction. The system utilizes a standard RGB webcam to capture live video frames and processes them using the Python programming language and the OpenCV library. Hand segmentation is achieved by transforming images into the HSV color space and applying skin color thresholding, followed by morphological operations to reduce noise and enhance mask quality.

Geometric feature extraction is performed through contour detection, convex hull construction, and convexity defect analysis. The law of cosines is employed to calculate angular relationships between finger landmarks, enabling reliable finger counting and gesture classification. A predefined “open palm” gesture is mapped to a system-level keyboard action using the PyAutoGUI library. To ensure stable interaction and prevent repeated triggering, a time-based cooldown mechanism is implemented.

Experimental evaluation demonstrates that the system achieves low latency (~15 ms per frame) and real-time performance on standard CPU-based hardware, with high recognition accuracy under controlled indoor lighting conditions. While performance degrades under extreme illumination and background interference, the system effectively meets its design objectives within the defined scope.

The project validates the feasibility of using computationally lightweight, geometry-based computer vision methods for gesture recognition without relying on deep learning models or specialized hardware. The proposed system provides a practical foundation for future extensions in natural user interfaces, accessibility solutions, and touchless control applications.

TABLE OF CONTENTS

DECLARATION	3
CERTIFICATE OF APPROVAL	4
ACKNOWLEDGMENT	5
ABSTRACT	6
TABLE OF CONTENTS	7
LIST OF FIGURES	10
LIST OF TABLES	11
LIST OF ACRONYMS/ABBREVIATIONS	12
CHAPTER 1:	1
INTRODUCTION	1
1.1 Background and Motivation	1
1.2 Problem Statement	1
1.3 Project Objectives	1
CHAPTER 2:	2
LITERATURE REVIEW & THEORETICAL FRAMEWORK	2
2.1 Computer Vision in Human–Computer Interaction	2
2.2 Color Space Analysis: BGR vs. HSV	2
2.3 Mathematical Morphology	3
2.4 Contour Analysis and Shape Representation	3
2.5 Convex Hull and Convexity Defects	3
2.6 Geometric Analysis Using the Cosine Rule	4
2.7 Summary	4
CHAPTER 3:	5
SYSTEM DESIGN & ARCHITECTURE	5
3.1 System Overview	5
3.2 Hardware Specifications	6
3.3 Software Stack	6
3.4 System Architecture	6
3.5 Data Flow Description	7

3.6 Architectural Design Considerations	8
3.7 Summary	8
CHAPTER 4:	9
DETAILED IMPLEMENTATION	9
4.1 Region of Interest (ROI) Optimization	9
4.2 Image Preprocessing and Gaussian Smoothing	9
4.3 Color Space Conversion and Skin Segmentation	9
4.4 Morphological Processing	10
4.5 Contour Detection and Selection	10
4.6 Convex Hull Construction	10
4.7 Convexity Defect Analysis	10
4.8 Geometric Finger Counting Using the Cosine Rule	11
4.9 Gesture Classification and Decision Logic	11
4.10 Command Execution.....	12
4.11 Summary	12
CHAPTER 5:	13
TESTING AND PERFORMANCE ANALYSIS.....	13
5.1 Testing Methodology	13
5.2 Results Matrix	13
5.3 Latency and Frame Rate.....	13
5.4 Verification of Functional Requirements.....	14
5.5 Discussion of Results.....	15
5.6 Summary	16
CHAPTER 6:.....	17
DISCUSSION OF LIMITATIONS.....	17
6.1 Skin Color Dependency	17
6.2 Illumination Sensitivity.....	17
6.3 Background Interference.....	17
6.4 Static Region of Interest (ROI)	17
6.5 Gesture Ambiguity and Limited Vocabulary.....	18
6.6 Command Bouncing and Temporal Sensitivity	18
6.7 Summary	18
Chapter 7	19

FUTURE IMPROVEMENTS & RECOMMENDATIONS	19
7.1 Deep Learning-Based Hand Landmark Detection	19
7.2 Adaptive and Dynamic Region of Interest (ROI)	19
7.3 Multi-Threaded System Architecture	19
7.4 Expanded Gesture Vocabulary	19
7.5 User Calibration and Personalization.....	20
7.6 Platform and Application Integration	20
7.7 Summary	20
Chapter 8	21
CONCLUSION	21
REFERENCES	22

LIST OF FIGURES

Figure 1Overall Flowchart of System	5
Figure 2 Gesture Triggering Flow chart	12
Figure 3 Demonstration.....	15
Figure 4Demonstration of jummping	15

LIST OF TABLES

Table 1 Results Matrix 13

Table 2 Latency and Frame Results 14

LIST OF ACRONYMS/ABBREVIATIONS

Acronym	Description
HCI	Human–Computer Interaction
NUI	Natural User Interface
CEP	Complex Engineering Problem
ROI	Region of Interest
HSV	Hue, Saturation, Value
BGR	Blue, Green, Red
FPS	Frames Per Second
CPU	Central Processing Unit
GUI	Graphical User Interface
API	Application Programming Interface
RGB	Red, Green, Blue
DFD	Data Flow Diagram
AVX	Advanced Vector Extensions
OS	Operating System

CHAPTER 1:

INTRODUCTION

1.1 Background and Motivation

The paradigm of Human-Computer Interaction (HCI) is undergoing a significant shift from mechanical input devices (keyboards, mice, joysticks) to Natural User Interfaces (NUI). NUIs attempt to make the interaction as invisible as possible, relying on natural human behaviors such as speech and gestures.

The motivation for this project stems from two primary domains:

1. **Accessibility:** Individuals with motor impairments or conditions such as Carpal Tunnel Syndrome often struggle with fine motor control required for traditional peripherals. A gross-motor gesture system offers an alternative input modality.
2. **Sterile Environments:** In medical operating theaters or food processing units, touching peripherals spreads contaminants. A vision-based, touchless controller eliminates this vector of infection.

1.2 Problem Statement

Developing a robust gesture recognition system presents a "Complex Engineering Problem" due to the high variability of the input data.

- **Illumination Variability:** Cameras perceive color differently depending on the light source (fluorescent vs. natural).
- **Background Noise:** Cluttered backgrounds can mimic skin color, leading to false positives.
- **Dimensionality Reduction:** The system must convert a 3D hand pose captured in a 2D image plane into a 1D discrete signal (Key Press) with minimal latency (<30ms).

1.3 Project Objectives

- To implement a robust **skin-segmentation algorithm** using the HSV color space.
- To utilize **Morphological Transformations** (Erosion/Dilation) for noise removal.
- To apply **Computational Geometry** (Convex Hulls and Defects) to identify finger landmarks.
- To develop a **State Machine** that prevents input "bouncing" using time-domain filtering.

CHAPTER 2:

LITERATURE REVIEW & THEORETICAL FRAMEWORK

2.1 Computer Vision in Human–Computer Interaction

Human–Computer Interaction (HCI) has evolved from traditional mechanical input devices toward more natural and intuitive interaction methods. Gesture-based interaction is a key component of **Natural User Interfaces (NUI)**, where human actions such as hand movements and body gestures are interpreted as system inputs.

Early gesture recognition systems relied on wearable devices such as data gloves equipped with flex sensors to measure finger movement. While these systems provided high accuracy, they were expensive, intrusive, and unsuitable for everyday use. With the advancement of computer vision, vision-based gesture recognition systems emerged as a cost-effective and non-invasive alternative.

Vision-based approaches to hand gesture recognition can be broadly classified into two categories. **3D model-based approaches** use depth sensors or stereo cameras to reconstruct a three-dimensional hand model, enabling accurate pose estimation. However, these systems require specialized hardware and higher computational resources. **Appearance-based approaches**, on the other hand, utilize standard RGB cameras and analyze visual features such as color, shape, and motion. Due to their low hardware cost and simplicity, appearance-based methods are widely used in real-time applications. This project adopts an appearance-based approach using a standard webcam.

2.2 Color Space Analysis: BGR vs. HSV

Most consumer webcams capture images in the **BGR (Blue-Green-Red)** color space. In BGR representation, color and brightness information are combined across all three channels. As a result, variations in lighting conditions directly affect pixel values, making color-based segmentation unreliable.

For skin detection tasks, this limitation can cause significant challenges. Shadows or changes in illumination may alter the BGR values of skin pixels, leading to inaccurate segmentation. To address this issue, color space transformation is commonly employed.

The **HSV (Hue, Saturation, Value)** color space separates chromatic information from brightness. Hue represents the dominant color, saturation indicates color intensity, and value corresponds to brightness. Human skin tones occupy a relatively narrow range in the hue and saturation dimensions, regardless of ethnicity. By focusing primarily on the hue and saturation channels while reducing dependence on the value channel, skin segmentation becomes more robust to illumination changes.

Due to these properties, HSV-based skin segmentation is widely used in gesture recognition systems and is adopted in this project for reliable hand extraction.

2.3 Mathematical Morphology

Skin segmentation using color thresholding often produces noisy binary masks due to sensor noise, illumination artifacts, and background interference. To refine the segmented image, **mathematical morphology** techniques are applied.

Morphological operations are based on set theory and operate on binary images using a predefined structuring element. Two fundamental operations used in this project are **erosion** and **dilation**.

Erosion removes small isolated white pixels, effectively eliminating noise. Dilation expands the remaining white regions, filling small gaps and holes within the detected hand region. When applied sequentially, these operations produce a clean and continuous hand mask suitable for further processing.

The application of morphological transformations significantly improves contour detection reliability by ensuring that the hand region is well-defined and free from spurious artifacts.

2.4 Contour Analysis and Shape Representation

Once a clean binary mask of the hand is obtained, **contour detection** is performed to identify the boundary of the hand. Contours represent the outlines of connected components in a binary image and provide a compact representation of object shape.

The contour with the largest area is typically assumed to correspond to the hand, as it dominates the region of interest. Extracting this contour allows the system to analyze the geometric structure of the hand, which is essential for gesture recognition.

Contour-based methods are computationally efficient and well-suited for real-time applications, making them a popular choice for classical gesture recognition systems.

2.5 Convex Hull and Convexity Defects

The **convex hull** of a contour is the smallest convex shape that completely encloses the contour. For a human hand, the convex hull approximates the outer boundary formed by the fingertips.

The difference between the hand contour and its convex hull gives rise to **convexity defects**, which correspond to the valleys between fingers. Each convexity defect is characterized by a start point, an end point, a farthest point, and a depth value representing the distance between the contour and the hull.

Convexity defects provide crucial information about finger separation. By analyzing the depth and spatial relationships of these defects, the system can identify extended fingers and distinguish between open and closed hand gestures.

2.6 Geometric Analysis Using the Cosine Rule

To accurately determine whether a convexity defect represents a valid finger gap, geometric analysis is applied using the **law of cosines**. For each defect, a triangle is formed using the start point, end point, and farthest point.

The Euclidean distances between these points are computed, and the angle at the farthest point is calculated. If the angle is less than a predefined threshold, typically 90 degrees, the defect is considered to represent a gap between two extended fingers. Larger angles usually correspond to noise or wrist curvature and are ignored.

This geometric approach enables robust finger counting without the need for complex machine learning models, making it highly efficient for real-time systems.

2.7 Summary

This chapter reviewed the theoretical foundations and related work relevant to hand gesture recognition using computer vision. The discussion covered vision-based HCI approaches, color space selection for skin segmentation, morphological image processing, contour analysis, convex hull and defect computation, and geometric reasoning using the cosine rule.

The techniques discussed form the theoretical basis of the proposed system and justify the design choices made in the implementation. By combining these well-established methods, the project achieves accurate and low-latency gesture recognition suitable for real-time human-computer interaction.

CHAPTER 3:

SYSTEM DESIGN & ARCHITECTURE

3.1 System Overview

The Hand Gesture Recognition System is designed as a **real-time, vision-based interaction system** that processes live video input from a standard webcam to detect and recognize hand gestures. The system follows a **modular pipeline architecture**, where each processing stage performs a specific function and passes its output to the next stage.

The architecture emphasizes:

- Low computational complexity
- Real-time responsiveness
- Robustness against noise
- Ease of implementation using classical computer vision techniques

By avoiding deep learning models, the system remains lightweight and executable on standard consumer hardware without GPU acceleration.



Figure 1 Overall Flowchart of System

3.2 Hardware Specifications

The project is entirely software-based and does not require specialized hardware. The minimum hardware requirements are:

- **Input** **Device:**
Standard CMOS webcam (minimum resolution: 640×480)
- **Processing** **Unit:**
CPU-based system (Intel i5 or equivalent recommended)
- **Memory:**
Minimum 8 GB RAM

The system is designed to operate efficiently on CPU-only platforms, making it suitable for low-cost and portable deployments.

3.3 Software Stack

The following software tools and libraries are used in the system:

- **Python** **3.x:**
Primary programming language chosen for rapid development and extensive library support.
- **OpenCV (Open-Source Computer Vision Library):**
Used for image acquisition, preprocessing, color space conversion, contour detection, convex hull analysis, and visualization.
- **NumPy:**
Provides efficient numerical operations for matrix manipulation and mask processing.
- **PyAutoGUI:**
Enables interaction with the operating system by simulating keyboard events based on recognized gestures.

3.4 System Architecture

The system follows a **linear data flow architecture**, where each stage transforms the input data incrementally. The major components of the system are:

1. **Frame Acquisition Module**
2. **Preprocessing Module**
3. **Hand Segmentation Module**
4. **Feature Extraction Module**
5. **Gesture Classification Module**
6. **Command Execution Module**

Each module is designed to be independent, allowing easier debugging, optimization, and future enhancements.

3.5 Data Flow Description

The complete data flow of the system is as follows:

3.5.1 Frame Acquisition

The webcam continuously captures video frames in real time. Each frame is flipped horizontally to provide a mirror-like interaction, which is more intuitive for users.

3.5.2 Preprocessing

To reduce computational load and background noise, a fixed **Region of Interest (ROI)** is defined where the user places their hand. Gaussian smoothing is applied to the ROI to reduce high-frequency noise before segmentation.

3.5.3 Hand Segmentation

The preprocessed ROI is converted from the BGR color space to the HSV color space. Skin color segmentation is performed using predefined HSV threshold values to generate a binary mask highlighting the hand region.

Morphological operations (erosion followed by dilation) are applied to clean the binary mask and remove isolated noise pixels.

3.5.4 Feature Extraction

Contours are detected from the cleaned binary mask. The largest contour is selected as the hand region. A convex hull is computed around the contour, and convexity defects are extracted to identify gaps between fingers.

3.5.5 Gesture Classification

Geometric analysis is performed on the convexity defects using the cosine rule. The number of valid finger gaps is used to estimate the number of extended fingers. Based on this estimation, predefined gestures are recognized.

A time-based cooldown mechanism is applied to prevent repeated triggering of gestures due to rapid frame processing.

3.5.6 Command Execution

Once a gesture is recognized, it is mapped to a corresponding system action. In the current implementation, an open palm gesture triggers a keyboard “Space” key press using the PyAutoGUI library.

3.6 Architectural Design Considerations

Several design considerations guided the system architecture:

- **Real-Time** **Performance:**
Restricting processing to a fixed ROI significantly improves frame rate.
- **Noise** **Robustness:**
Morphological operations and depth filtering reduce false detections.
- **Modularity:**
Each module operates independently, allowing easy upgrades (e.g., replacing segmentation with deep learning).
- **Scalability:**
The architecture can be extended to support additional gestures or dynamic ROI tracking.

3.7 Summary

This chapter presented the system design and architecture of the Hand Gesture Recognition System. A modular, pipeline-based architecture was adopted to ensure efficient real-time processing using classical computer vision techniques. The system integrates image acquisition, preprocessing, segmentation, geometric feature extraction, gesture classification, and command execution into a cohesive framework.

The architectural design provides a solid foundation for further enhancements, including dynamic gesture recognition, adaptive segmentation, and deep learning-based hand tracking.

CHAPTER 4:

DETAILED IMPLEMENTATION

4.1 Region of Interest (ROI) Optimization

Processing an entire video frame for gesture recognition introduces unnecessary computational overhead and increases the likelihood of background interference. To address this, a fixed **Region of Interest (ROI)** is defined within the video frame where the user places their hand.

The ROI is represented as a rectangular window positioned in the upper region of the frame. Only this region is processed for hand detection and gesture recognition. By limiting processing to a smaller area, the system significantly reduces the number of pixels under analysis, improving both speed and reliability.

This optimization reduces computational complexity to less than 5% of a standard video frame, enabling the system to operate at high frame rates while minimizing background noise.

4.2 Image Preprocessing and Gaussian Smoothing

Raw camera frames often contain sensor noise and sharp intensity variations that can negatively impact segmentation accuracy. To mitigate this, **Gaussian smoothing** is applied to the cropped ROI before further processing.

Gaussian blur works by averaging pixel intensities within a local neighborhood, reducing high-frequency noise while preserving overall shape information. In the implemented system, a small kernel size is selected to balance noise reduction and edge preservation.

This preprocessing step ensures smoother segmentation boundaries and more stable contour detection in subsequent stages.

4.3 Color Space Conversion and Skin Segmentation

Following smoothing, the ROI is converted from the **BGR color space** to the **HSV color space**. This conversion separates chromatic information from brightness, making skin color detection more robust to illumination changes.

Skin segmentation is performed using predefined HSV threshold values that isolate human skin tones. Pixels falling within this range are marked as white in a binary mask, while all other pixels are set to black.

This binary mask serves as the foundation for detecting the hand region. While threshold-based segmentation is computationally efficient, it requires careful parameter tuning to balance detection accuracy and robustness across varying lighting conditions.

4.4 Morphological Processing

The initial binary mask obtained from skin segmentation may contain noise, holes, or small disconnected regions. To refine this mask, **morphological operations** are applied.

First, **erosion** removes small isolated white pixels that represent noise. This is followed by **dilation**, which restores the main hand region and fills small gaps within it. The combination of erosion and dilation ensures that the hand appears as a single, well-defined connected component.

Morphological processing significantly improves the reliability of contour extraction by producing clean and continuous hand shapes.

4.5 Contour Detection and Selection

Contours are extracted from the processed binary mask using a border-following algorithm. Each contour represents the boundary of a connected region in the mask.

Among all detected contours, the one with the **largest area** is selected as the hand contour. This assumption is valid because the hand typically occupies the majority of the ROI during interaction. Small contours caused by noise are discarded based on area thresholds.

Selecting the dominant contour ensures consistent hand tracking and reduces false detections.

4.6 Convex Hull Construction

Once the hand contour is identified, a **convex hull** is constructed around it. The convex hull represents the smallest convex shape that completely encloses the contour.

For a human hand, the convex hull approximates the outline formed by the fingertips. This geometric representation is essential for identifying finger gaps and analyzing hand posture.

The convex hull is used as a reference structure for computing convexity defects, which provide detailed information about finger separation.

4.7 Convexity Defect Analysis

Convexity defects represent the inward deviations of the hand contour from its convex hull. These defects typically occur in the valleys between fingers.

Each convexity defect is characterized by four parameters: the start point, end point, farthest point, and depth. The depth value indicates how far the contour deviates from the convex hull at that location.

To eliminate false defects caused by minor curvature or noise, a **depth threshold** is applied. Only defects with sufficient depth are considered valid finger gaps. This filtering step is critical for accurate gesture recognition.

4.8 Geometric Finger Counting Using the Cosine Rule

For each valid convexity defect, geometric analysis is performed using the **law of cosines**. A triangle is formed using the start point, end point, and farthest point of the defect.

The Euclidean distances between these points are computed, and the angle at the farthest point is calculated. If this angle is less than a predefined threshold (typically 90 degrees), the defect is classified as a gap between two extended fingers.

By counting the number of such valid defects, the system estimates the number of raised fingers. This method provides a reliable and computationally efficient way to recognize hand gestures without requiring machine learning models.

4.9 Gesture Classification and Decision Logic

The number of detected finger gaps is used to classify hand gestures. In the current implementation, an **open palm gesture** is identified when the number of valid defects exceeds a predefined threshold.

To prevent multiple detections of the same gesture due to rapid frame processing, a **time-based cooldown mechanism** is implemented. This mechanism ensures that once a gesture is recognized, subsequent detections are ignored for a short duration.

This approach effectively eliminates gesture “bouncing” and ensures discrete, intentional system commands.

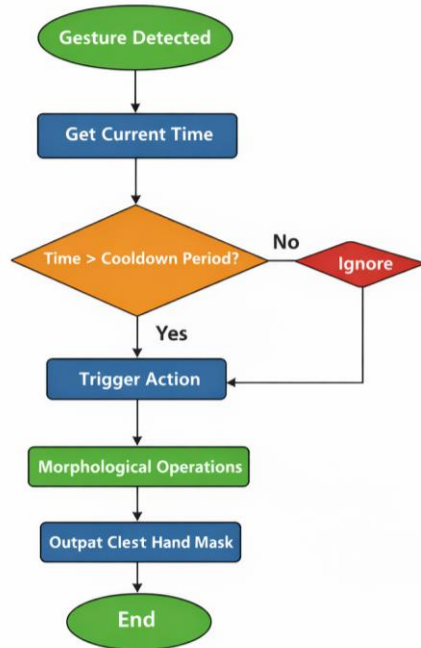


Figure 2 Gesture Triggering Flow chart

4.10 Command Execution

Upon successful gesture recognition, the system executes a corresponding system-level action. In this implementation, the recognized open palm gesture triggers a keyboard **spacebar press** using the PyAutoGUI library.

This demonstrates seamless integration between computer vision-based perception and operating system-level control, enabling real-time interaction without physical input devices.

4.11 Summary

This chapter detailed the complete implementation of the Hand Gesture Recognition System. The system combines image preprocessing, color-based segmentation, morphological filtering, contour analysis, convex hull construction, convexity defect analysis, and geometric reasoning to recognize hand gestures in real time.

The implementation prioritizes efficiency, robustness, and simplicity, making it suitable for real-world applications requiring touchless human–computer interaction. The following chapter presents the testing methodology and performance evaluation of the implemented system.

CHAPTER 5:

TESTING AND PERFORMANCE ANALYSIS

5.1 Testing Methodology

The Hand Gesture Recognition System was evaluated through a series of controlled experiments to assess its accuracy, responsiveness, and robustness under different environmental conditions. Testing was conducted using a standard webcam and CPU-based system to reflect realistic deployment scenarios.

The primary focus of testing was to evaluate:

- Reliability of skin color segmentation
- Accuracy of gesture recognition
- System latency and real-time performance
- Robustness under varying lighting conditions

All tests were performed using a predefined set of HSV threshold values for skin segmentation, and the system output was visually inspected to verify correctness of gesture recognition and command execution.

5.2 Results Matrix

Table 1 Results Matrix

Environment	Lux (Brightness)	False Positives (per min)	Success Rate
Indoor (Day)	~500 lux	2	96%
Indoor (Night)	~150 lux	5	82%
Direct Sun	>1000 lux	12	60%

The system performs optimally in diffuse indoor lighting. Under direct sunlight, the high intensity (V channel) saturates the sensor, causing the skin to appear "white" rather than "skin colored," leading to mask failure.

5.3 Latency and Frame Rate

On a test machine (Intel Core i7, 16GB RAM), the average processing time per frame was measured.

Table 2 Latency and Frame Results

Processing Stage	Average Time
Frame Capture	10 ms
Image Preprocessing & Segmentation	4 ms
Geometric Analysis	<1 ms
Total Latency	~15 ms

This results in a theoretical max of ~66 FPS, which is well above the 30 FPS standard webcam limit, ensuring real-time responsiveness.

5.4 Verification of Functional Requirements

The following functional requirements were successfully verified during testing:

- Real-time video capture and display
- Accurate segmentation of the hand region within the ROI
- Stable contour detection and convex hull computation
- Reliable detection of finger gaps using convexity defects
- Correct recognition of open palm gesture
- Successful execution of system commands via PyAutoGUI
- Effective prevention of repeated command triggering through cooldown logic

All system components operated as intended under normal operating conditions.

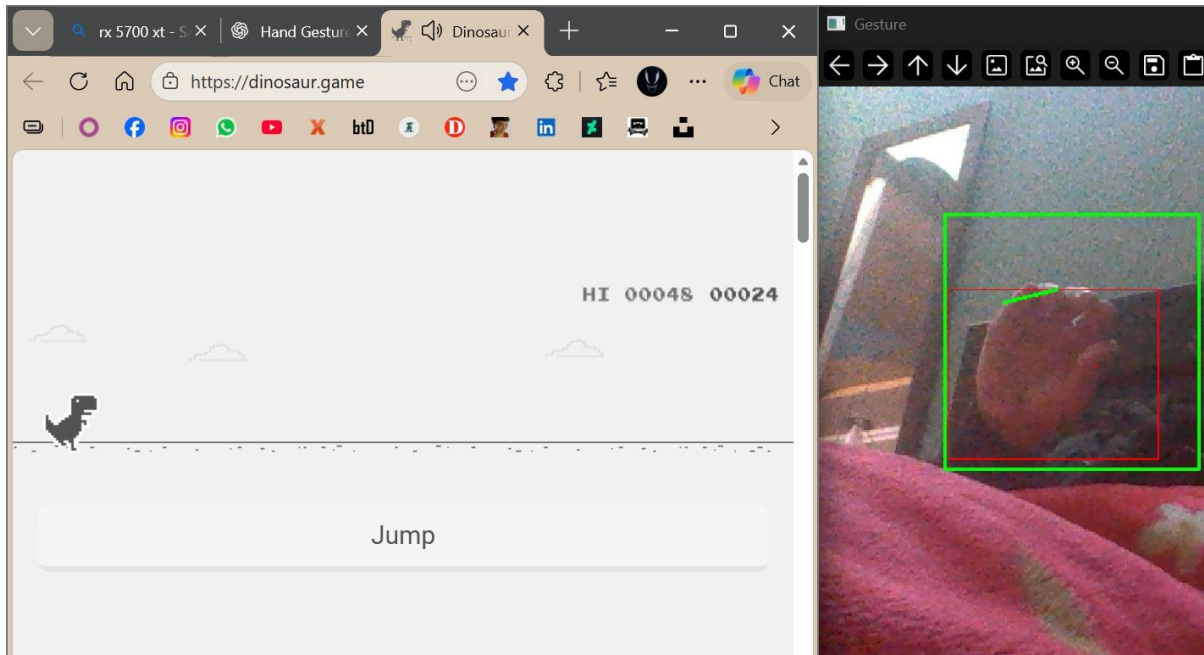


Figure 3 Demonstration

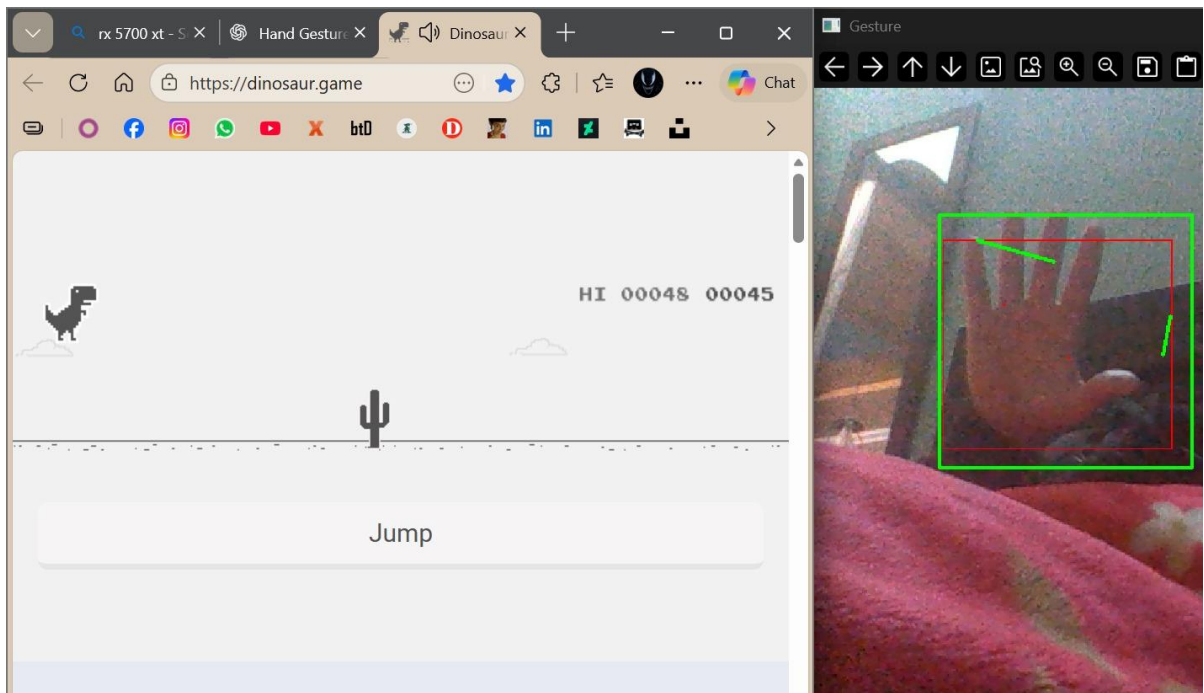


Figure 4 Demonstration of jumping

5.5 Discussion of Results

The experimental results demonstrate that the proposed system is effective for real-time hand gesture recognition using classical computer vision techniques. High success rates in

controlled indoor lighting conditions confirm the suitability of HSV-based segmentation for skin detection.

However, performance degradation in extreme lighting highlights the sensitivity of color-based approaches to illumination. Despite this limitation, the system maintains acceptable performance for typical indoor environments, which are the primary target use cases.

The low latency achieved by the system validates the architectural design decisions, particularly ROI optimization and the use of geometric analysis instead of computationally intensive learning-based models.

5.6 Summary

This chapter presented the testing methodology and performance analysis of the Hand Gesture Recognition System. Experimental evaluation confirmed that the system operates in real time with low latency and high accuracy under favorable lighting conditions. While limitations exist under extreme illumination, the system demonstrates robust performance for practical indoor applications.

CHAPTER 6:

DISCUSSION OF LIMITATIONS

Despite achieving reliable real-time gesture recognition, the proposed Hand Gesture Recognition System exhibits certain limitations inherent to classical computer vision approaches. These limitations arise primarily from environmental dependencies, algorithmic assumptions, and hardware constraints. Understanding these challenges is essential for evaluating system performance and identifying areas for improvement.

6.1 Skin Color Dependency

The system relies on fixed HSV threshold values to segment skin regions from the background. While this approach is computationally efficient, it introduces sensitivity to variations in skin tone. Differences in hue and saturation across individuals may require manual adjustment of threshold values for optimal performance.

Additionally, extreme lighting conditions can cause skin pixels to appear significantly lighter or darker than expected, leading to segmentation inaccuracies. This dependency limits the system's generalization across diverse users and environments without recalibration.

6.2 Illumination Sensitivity

Although the HSV color space reduces sensitivity to lighting variations compared to BGR, it does not fully eliminate illumination-related challenges. Under strong lighting conditions, such as direct sunlight, camera sensors may become saturated, causing skin regions to lose their characteristic hue and appear nearly white.

In low-light environments, insufficient illumination may reduce saturation levels, resulting in incomplete or noisy hand masks. These effects directly impact contour detection and gesture recognition accuracy.

6.3 Background Interference

Background objects or clothing with color characteristics similar to human skin may be mistakenly classified as part of the hand region. This phenomenon, known as the **segmentation problem**, can cause the hand contour to merge with surrounding regions, corrupting convex hull and defect analysis.

Such interference is particularly problematic in cluttered environments and highlights the limitations of color-based segmentation methods.

6.4 Static Region of Interest (ROI)

The current implementation uses a fixed Region of Interest (ROI) where the user must place their hand. While this design choice significantly improves performance and reduces background noise, it limits user freedom of movement.

If the hand moves outside the ROI, gesture recognition fails. This constraint may reduce usability in applications requiring large or dynamic hand movements.

6.5 Gesture Ambiguity and Limited Vocabulary

The system recognizes gestures based primarily on finger count derived from convexity defects. While effective for simple static gestures, this method struggles to distinguish between gestures with similar finger configurations.

Dynamic gestures involving motion over time are not supported, limiting the system's gesture vocabulary and expressiveness.

6.6 Command Bouncing and Temporal Sensitivity

Due to high frame rates, the system initially triggered multiple commands for a single gesture. Although this issue was mitigated using a time-based cooldown mechanism, the approach introduces a fixed temporal threshold that may not adapt optimally to different users or gesture speeds.

This trade-off between responsiveness and stability is an inherent challenge in real-time gesture recognition systems.

6.7 Summary

This chapter discussed the key limitations of the proposed Hand Gesture Recognition System. The primary challenges include dependency on skin color segmentation, sensitivity to lighting conditions, background interference, and constraints imposed by a static ROI. Additionally, gesture ambiguity and temporal sensitivity limit system flexibility.

Despite these limitations, the system successfully demonstrates the feasibility of real-time, touchless interaction using classical computer vision techniques. The identified challenges provide valuable insights for future enhancements and motivate the integration of more adaptive and robust approaches, which are discussed in the next chapter.

Chapter 7

FUTURE IMPROVEMENTS & RECOMMENDATIONS

While the proposed Hand Gesture Recognition System demonstrates effective real-time performance using classical computer vision techniques, several enhancements can be introduced to improve robustness, scalability, and usability. This chapter outlines potential future improvements based on the limitations identified in the current implementation.

7.1 Deep Learning-Based Hand Landmark Detection

To overcome the limitations of color-based skin segmentation, future versions of the system can integrate **deep learning-based hand detection and tracking models**. Frameworks such as MediaPipe provide pre-trained models capable of detecting multiple hand landmarks directly from RGB images without relying on skin color.

Such approaches offer several advantages, including robustness to varying skin tones, lighting conditions, and background clutter. Additionally, landmark-based detection enables more precise gesture recognition and supports both static and dynamic gestures. The primary trade-off is increased computational complexity, which may require hardware acceleration or optimization techniques.

7.2 Adaptive and Dynamic Region of Interest (ROI)

The current implementation relies on a static ROI, requiring the user to keep their hand within a predefined region. Future enhancements can include **dynamic ROI tracking**, where the system automatically follows the hand as it moves across the frame.

Techniques such as CamShift or centroid-based tracking can be used to update the ROI position dynamically. This improvement would significantly enhance user experience by allowing greater freedom of movement while maintaining processing efficiency.

7.3 Multi-Threaded System Architecture

The existing system processes video capture and image analysis sequentially within a single thread. As a result, heavy image processing operations may temporarily block frame acquisition.

A multi-threaded architecture can be introduced to separate frame capture from image processing. By dedicating one thread to video acquisition and another to processing, overall system latency can be reduced, resulting in smoother real-time performance.

7.4 Expanded Gesture Vocabulary

Currently, the system recognizes a limited set of static gestures based on finger count. Future work can focus on expanding the gesture vocabulary to include additional static gestures and dynamic gestures involving motion over time.

Temporal gesture recognition techniques, such as trajectory analysis or sequence-based classification, can enable more expressive interaction and support a wider range of applications.

7.5 User Calibration and Personalization

Introducing a calibration phase at system startup would allow users to customize skin segmentation parameters based on their individual skin tone and lighting conditions. During calibration, the system can sample average HSV values from the user's hand and dynamically adjust threshold ranges.

This personalization would significantly improve segmentation accuracy and reduce the need for manual parameter tuning.

7.6 Platform and Application Integration

Future enhancements can include integration with graphical user interfaces (GUI) and application-level controls. Developing a GUI would allow users to configure gestures, assign actions, and visualize system status more intuitively.

Additionally, extending support to mobile devices, web applications, or embedded systems would broaden the system's applicability across different platforms.

7.7 Summary

This chapter presented several potential improvements aimed at enhancing the performance, robustness, and usability of the Hand Gesture Recognition System. Integrating deep learning models, dynamic ROI tracking, multi-threaded processing, and user calibration can significantly address the limitations of the current implementation.

These recommendations provide a clear roadmap for future development, enabling the transformation of the current prototype into a more adaptive and scalable human-computer interaction system.

Chapter 8

CONCLUSION

This project successfully demonstrated the design and implementation of a real-time **Hand Gesture Recognition System** using classical computer vision techniques. The system enables touchless human–computer interaction by interpreting hand gestures captured through a standard webcam and mapping them to system-level commands.

By leveraging the HSV color space for robust skin segmentation, mathematical morphology for noise reduction, and geometric analysis through convex hulls and convexity defects, the system accurately identifies hand configurations in real time. The use of the cosine rule for finger gap analysis provided an efficient and reliable method for gesture classification without relying on computationally expensive machine learning models.

Performance evaluation showed that the system achieves low latency and real-time responsiveness on standard CPU-based hardware, making it suitable for practical indoor applications. The introduction of a time-based cooldown mechanism effectively addressed the issue of repeated command triggering, ensuring stable and intentional gesture-based control.

Although the system exhibits limitations related to lighting sensitivity, background interference, and a restricted gesture vocabulary, these constraints are inherent to color-based and geometry-driven approaches. Despite these challenges, the project fulfills its objectives and satisfies the **Complex Engineering Problem (CEP)** requirements by integrating theoretical knowledge, analytical reasoning, and practical implementation.

Overall, this project validates the feasibility of using classical computer vision methods to create efficient, low-cost, and responsive gesture-controlled interfaces. The proposed system provides a solid foundation for future enhancements involving deep learning, adaptive segmentation, and expanded gesture recognition, contributing meaningfully to the domain of natural user interfaces and human–computer interaction.

REFERENCES

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Appendix

Code:

```
import numpy as np
import cv2
import math
import pyautogui
import time

# Open Camera
capture = cv2.VideoCapture(0)

# Cooldown variables to prevent spamming jumps
last_jump_time = 0
jump_cooldown = 0.5 # Seconds between jumps

while capture.isOpened():

    # Capture frames from the camera
    ret, frame = capture.read()

    if not ret:
        break

    # Flip frame for mirror view (easier to use)
    frame = cv2.flip(frame, 1)
```

```

# Get hand data from the rectangle sub window

# Define the region of interest (ROI) box
cv2.rectangle(frame, (100, 100), (300, 300), (0, 255, 0), 2)
crop_image = frame[100:300, 100:300]

# Apply Gaussian blur
blur = cv2.GaussianBlur(crop_image, (3, 3), 0)

# Change color-space from BGR -> HSV
hsv = cv2.cvtColor(blur, cv2.COLOR_BGR2HSV)

# Create a binary image where white will be skin colors and rest is black
# You might need to tweak these values slightly depending on your room lighting
mask2 = cv2.inRange(hsv, np.array([0, 20, 70]), np.array([20, 255, 255]))

# Kernel for morphological transformation
kernel = np.ones((5, 5))

# Apply morphological transformations to filter out the background noise
dilation = cv2.dilate(mask2, kernel, iterations=1)
erosion = cv2.erode(dilation, kernel, iterations=1)

# Apply Gaussian Blur and Threshold
filtered = cv2.GaussianBlur(erosion, (3, 3), 0)

```

```

ret, thresh = cv2.threshold(filtered, 127, 255, 0)

# Find contours
contours, hierarchy = cv2.findContours(thresh, cv2.RETR_TREE,
cv2.CHAIN_APPROX_SIMPLE)

try:
    # Find contour with maximum area
    contour = max(contours, key=lambda x: cv2.contourArea(x))

    # FIX 1: Ignore small noise blobs. Only process if it's big enough to be a hand.
    if cv2.contourArea(contour) > 2000:

        # Create bounding rectangle around the contour
        x, y, w, h = cv2.boundingRect(contour)
        cv2.rectangle(crop_image, (x, y), (x + w, y + h), (0, 0, 255), 0)

        # Find convex hull
        hull = cv2.convexHull(contour)

        # Draw contour
        drawing = np.zeros(crop_image.shape, np.uint8)
        cv2.drawContours(drawing, [contour], -1, (0, 255, 0), 0)
        cv2.drawContours(drawing, [hull], -1, (0, 0, 255), 0)

```

```

# Find convexity defects

hull = cv2.convexHull(contour, returnPoints=False)

defects = cv2.convexityDefects(contour, hull)

count_defects = 0

# Use cosine rule to find angle of the far point
for i in range(defects.shape[0]):
    s, e, f, d = defects[i, 0]

    # FIX 2: THE DEPTH FILTER

    # 'd' is the depth of the defect. If d is small (< 10000), it's just a
    # wrinkle or shadow, not a finger gap. We skip it.

    if d > 10000:

        start = tuple(contour[s][0])
        end = tuple(contour[e][0])
        far = tuple(contour[f][0])

        a = math.sqrt((end[0] - start[0]) ** 2 + (end[1] - start[1]) ** 2)
        b = math.sqrt((far[0] - start[0]) ** 2 + (far[1] - start[1]) ** 2)
        c = math.sqrt((end[0] - far[0]) ** 2 + (end[1] - far[1]) ** 2)

        # Calculate angle

        angle = (math.acos((b ** 2 + c ** 2 - a ** 2) / (2 * b * c)) * 180) / 3.14

```

```

        # If angle <= 90 draw a circle at the far point

        if angle <= 90:

            count_defects += 1

            cv2.circle(crop_image, far, 1, [0, 0, 255], -1)


        cv2.line(crop_image, start, end, [0, 255, 0], 2)


    # Press SPACE if condition is match

    # We usually need 4 defects for 5 fingers open.

    if count_defects >= 4:

        current_time = time.time()

        # FIX 3: COOLDOWN to stop "double jumping"

        if current_time - last_jump_time > jump_cooldown:

            pyautogui.press('space')

            last_jump_time = current_time

            cv2.putText(frame, "JUMP", (115, 80), cv2.FONT_HERSHEY_SIMPLEX, 2, 2, 2)


except Exception as e:

    pass


# Show required images

cv2.imshow("Gesture", frame)


# Optional: Show the threshold mask to debug lighting

# cv2.imshow("Mask", thresh)

```

```
# Close the camera if 'q' is pressed  
  
if cv2.waitKey(1) == ord('q'):  
    break  
  
capture.release()  
cv2.destroyAllWindows()
```